

Homework 4 – Biological Oceanography

Total 50 points

Due date: Tuesday 12/18/2025 at 10:45 am

Rules:

- Group assignment: 3-4 students (feel free to form new groups)
- Individual assignments will not be accepted
- ALL Answers MUST be organized in a single .docx or pdf file
- File MUST be submitted via email.
- In this email, all group members MUST be listed and copied
- Email MUST be sent to Instructure and BOTH TAs
- Extensions WILL NOT be granted
- If you do use generative AI, include a brief (one sentence) description of how you used the tool followed by a list of the prompts used (bullet points) used in each question
- IF THE GROUP FAILS TO SOLVE THE GROUP ACCOUNTABILITY EXERCISE, THE ENTIRE GROUP WILL RECEIVE A TOTAL OF 0 POINTS ON THE ENTIRE HOMEWORK ASSIGNMENT

Remember: All questions are supposed to be answered in the context of our classes. Even questions based on reading materials. Meaning, you will need fundamental knowledge presented in class together with the content presented here or reading materials.

- 1- Explain as detailed as possible how energy is captured (enters), stored, and transported through biological communities (10 pts)
- 2- Choose one of the animals cited during our final day of lecture and answer (10 Pts)
 - a. What is the common and scientific names of the chosen species? (2.5 pts)
 - b. List its COMPLETE taxonomic classification and what are the characteristics of each classification group (2.5 pts)
 - c. What type of marine community does this organism belong to and describe its geographic (2.5 pts)
 - d. What is the niche of this animal in the ecosystems it belongs to? (2.5 pts)
- 3- Read the article from Clark (2018) <https://www.whoi.edu/oceanus/feature/the-recipe-for-a-harmful-algal-bloom/> and answer (15 pts)
 - a. Explain why predicting *Pseudo-nitzschia* blooms are important and extremely challenging? (5 Pts)
 - b. How scientists use complex and numerous environmental observations to predict HABs? (5 Pts)

- c. If *Pseudo-nitzschia* blooms were only a function of the ocean temperature, what do you predict will happen to the frequency and intensity of HABs as a consequence of global warming? (5 Pts)
- 4- Read the article from Karl et al., (1995) and answer (15 Pts)
- The authors say that ocean gyres are like “ocean deserts”. How can these “deserts” significantly contribute to the global biological productivity? (2.5 Pts)
 - Based on the Abstract and Table 1, which ocean changes impact the North Pacific primary production when a strong El Niño event happens? (2.5 Pts)
 - Based on Table 1, which trends can be identified in the data when comparing periods with and without El Niño? Cite the numbers presented in the article that supports your argument (10 Pts)
- 5- **[MANDATORY] Group Accountability Exercise: In a short paragraph or a statement, describe the role of each group member during the homework assignment.**

ANY work done will be rewarded with a grade. Students with larger workloads will be rewarded extra points. Students with NO CONTRIBUTION to the group activity won't receive any points.

Example 1: All group members equally contributed to the homework assignment.

Example 2: Jane Doe answered questions 1 and 2, while John Doe, Jim Doe, and Carry Doe equally contributed to questions 3, 4, and 5.

Example 3: Everyone contributed except John Doe.

In example 2 Jane Doe will receive extra 5 points. In example 3, John Doe will fail the assignment (0 pts).

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